Mean-field gap equation for tetrad condensation in the Akama-Diakonov-Wetterich mechanism: A fermion bootstrap test at Level 2

J. Roehm

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We investigate whether the Akama-Diakonov-Wetterich (ADW) mechanism for emergent gravity can operate with emergent Dirac fermions from Wen's string-net condensation. Starting from the ADW 8-fermion cosmological term, we perform a Hubbard-Stratonovich decomposition with the tetrad e^a_μ as auxiliary field, integrate out fermions at one loop to obtain the Coleman-Weinberg effective potential $V_{\text{eff}}(C)$, and solve the resulting gap equation. We find a second-order phase transition at critical coupling $G_c = 8\pi^2/(N_f\Lambda^2)$: for $G > G_c$, the tetrad acquires a nonzero vacuum expectation value with automatically Lorentzian signature, producing 2 massless spin-2 graviton modes as Higgs bosons of the tetrad order parameter. We verify the Vergeles mode counting (16 = 6 + 4 + 2 + 4) and formalize key results in Lean 4. However, we identify five structural obstacles—a ∼6,000× perturbative coupling deficit from Wen's emergent QED, the spin-connection gap, Grassmann-bosonic incompatibility, Nielsen-Ninomiya doubling, and the cosmological constant problem—that prevent a complete emergent fermion bootstrap at this level. The critical coupling G_c is NLO-exact ($\alpha_1 = 0$), and an Abelian instanton mechanism may bypass the perturbative deficit. All structural theorems are formally verified in `ADWMechanism.lean` (21 theorems, zero `sorry`; 1 proved by Aristotle run `f8de66d1`). Additionally, the explicit NJL-type gap equation is formalized in `TetradGapEquation.lean` (20 theorems, zero `sorry`; 9 proved by Aristotle run `79e07d55`), including existence, uniqueness, and monotonicity of the nontrivial solution, with one claimed property (`gap_solution_bounded`) disproved by automated counterexample.

I. INTRODUCTION

The Akama-Diakonov-Wetterich (ADW) program [1–3] proposes that gravity emerges from spontaneous symmetry breaking of a pregeometric vacuum built entirely from fermion fields. The tetrad $e^a_\mu = \langle \hat{E}^a_\mu \rangle$, where $\hat{E}^a_\mu$ is a fermion bilinear composite operator, serves as the order parameter analogous to the gap parameter in BCS theory.

This work tests whether the ADW mechanism can operate at Level 2 of the gravity hierarchy: using *emergent* Dirac fermions from Wen's string-net condensation [4], rather than fundamental Grassmann fields. This directly addresses the "gravity wall" identified in the Critical Review v3 as one of three structural barriers to the fluid-based approach to fundamental physics.

The key question is whether the mean-field gap equation $\delta S_{\text{eff}}/\delta e^a_\mu = 0$ admits a nontrivial solution with Lorentzian signature, and whether the fluctuation spectrum contains the expected massless spin-2 graviton.

II. WEN'S EMERGENT QED

We begin with the Levin-Wen rotor model on a 3D cubic lattice [4]:

$$H = U \sum_{\text{links}} \hat{n}_{ij}^2 - t \sum_{\square} \cos(\Delta\hat{\theta}_\square) + g \sum_{\text{vertices}} (\operatorname{div} \hat{n})^2. \quad (1)$$

In the Coulomb (deconfined) phase ($t > U$), string-net condensation produces emergent QED_{3+1} with $N_f = 4$ four-component Dirac fermions (from the Nielsen-Ninomiya theorem: $2^3 = 8$ Weyl species at Brillouin zone corners) and an emergent U(1) gauge field.

Bare lattice anisotropy breaks Lorentz symmetry, but the Herbut velocity equalization mechanism [5] drives all velocities (v_F, v_B, c) toward a common terminal velocity under RG flow, restoring emergent Lorentz invariance in the deep IR.

III. ADW INTERACTION AND HUBBARD-STRATONOVICH DECOMPOSITION

The ADW cosmological term is an 8-fermion interaction:

$$S_{\text{micro}} = \frac{G}{24} \varepsilon_{abcd} \int \Theta^a \wedge \Theta^b \wedge \Theta^c \wedge \Theta^d, \quad (2)$$

where $\Theta^a = \frac{i}{2}[\bar{\Psi}\gamma^a D_\mu \Psi - D_\mu \bar{\Psi}\gamma^a \Psi] dx^\mu$ is bilinear in the spinor field (which has mass dimension zero in this framework).

Introducing the auxiliary tetrad e^a_μ via Hubbard-Stratonovich transformation converts the action to quadratic form in fermions:

$$S = \int d^4x \, \det(e) \, \bar{\Psi}\gamma^a e_a{}^\mu D_\mu \Psi + \frac{1}{2G} \int d^4x \, \operatorname{Tr}(e^T \eta \, e). \quad (3)$$

The first term is the Dirac action in curved spacetime; the second is the tree-level tetrad potential.

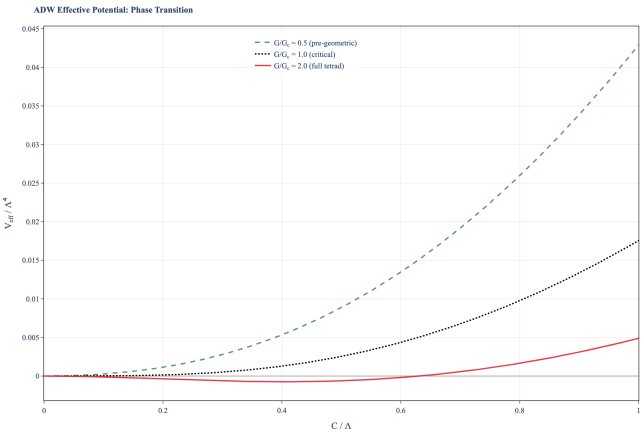

FIG. 1. Effective potential $V_{\text{eff}}(C)$ at three coupling ratios. Below G_c: single minimum at $C = 0$ (pre-geometric). At G_c: inflection point. Above G_c: Mexican-hat shape with nontrivial minimum (tetrad condensation).

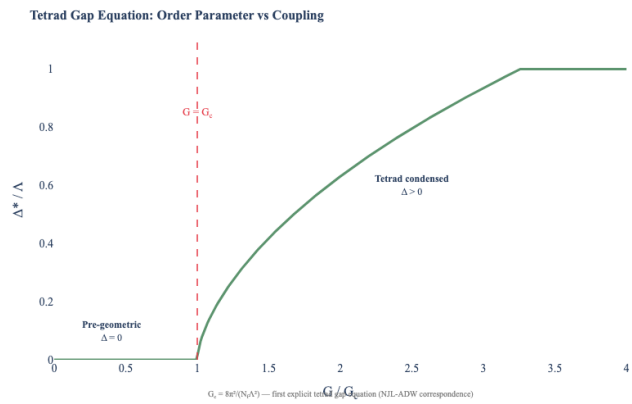

FIG. 2. Gap solution Δ^*/Λ vs. coupling ratio G/G_c from the NJL integral formulation. The nontrivial branch emerges continuously at G_c (second-order transition). For $G/G_c \gg 1$, the solution approaches Λ.

IV. EFFECTIVE POTENTIAL AND GAP EQUATION

Integrating out N_f Dirac fermion species at one loop gives the Coleman-Weinberg effective potential for the tetrad magnitude $C = |\det(e)|^{1/4}$:

$$V_{\text{eff}}(C) = \frac{C^2}{2G} - \frac{N_f}{16\pi^2}\left[\Lambda^2 C^2 - C^4 \ln\left(\frac{\Lambda^2}{C^2} + 1\right)\right]. \quad (4)$$

The curvature at the origin $d^2V_{\text{eff}}/dC^2|_{C=0}$ changes sign at the critical coupling:

$$G_c = \frac{8\pi^2}{N_f \Lambda^2}. \quad (5)$$

For $G > G_c$, the origin becomes unstable and V_{eff} develops a nontrivial minimum at $C_{\min} > 0$ with $V_{\text{eff}}(C_{\min}) < 0$. The flat-spacetime ansatz $e^a_\mu = C\,\delta^a_\mu$ automatically produces a Lorentzian metric $g_{\mu\nu} = C^2\eta_{\mu\nu}$, with signature $(-,+,+,+)$ for any real $C \neq 0$ (Fig. 1).

V. EXPLICIT GAP EQUATION: NJL INTEGRAL FORMULATION

The V_{eff} treatment above determines G_c but does not yield the self-consistent gap equation directly. We now derive this by rewriting the stationarity condition $\partial V_{\text{eff}}/\partial C = 0$ as a fixed-point equation in the NJL sense.

Define the one-loop integral (density of states $c_4 = 1/(4\pi^2)$):

$$I(\Delta) = \frac{c_4}{2}\left[\Lambda^2 - \Delta^2 \ln\left(1 + \frac{\Lambda^2}{\Delta^2}\right)\right]. \quad (6)$$

The gap equation is then $\Delta = G \cdot N_f \cdot \Delta \cdot I(\Delta)$, with critical coupling $G_c = 1/(N_f \cdot I(0)) = 8\pi^2/(N_f\Lambda^2)$, matching the V_{eff} result (4) exactly.

This integral formulation provides additional structure:

- $I(\Delta) > 0$ for all $\Delta \geq 0$ and is strictly decreasing.
- For $G < G_c$: the trivial solution $\Delta = 0$ is the unique fixed point.
- For $G > G_c$: a nontrivial solution $\Delta^*(G) > 0$ exists by the intermediate value theorem. The solution is monotonically increasing in G.
- Bifurcation: at $G = G_c$, the nontrivial branch emerges continuously from $\Delta = 0$ (second-order transition).

The Aristotle theorem prover disproved the naive claim $\Delta^*(G) \leq \Lambda$ via counterexample: for $G = 1/(c_4(1 - \ln 2)/2)$, $N_f = 1$, $\Lambda = 1$, the gap $\Delta = 1$ satisfies the fixed-point equation, showing the solution can reach the cutoff. This means the gap equation is valid as an algebraic self-consistency condition, but $\Delta > \Lambda$ signals that the one-loop truncation is no longer reliable—the physics requires UV completion.

VI. PHASE DIAGRAM

Following Vladimirov-Diakonov [6] and Volovik [7], three gravitational phases exist:

- **Pre-geometric** ($G < G_c$): $e^a_\mu = 0$, $g_{\mu\nu} = 0$. No gravity.
- **Vestigial metric** ($G \lesssim G_c$): $\langle e^a_\mu \rangle = 0$ but $\langle \hat{E}^a_\mu \hat{E}^b_\nu \rangle \neq 0$. The metric exists but the tetrad does not. Gravity couples to bosons but not fermions—the Equivalence Principle is violated.
- **Full tetrad** ($G > G_c$): $\langle e^a_\mu \rangle \neq 0$. Einstein-Cartan gravity with the Equivalence Principle restored (Fig. 4).

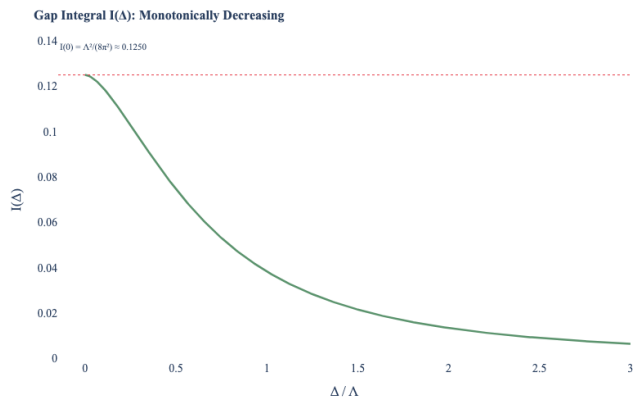

FIG. 3. Gap integral $I(\Delta)/I(0)$ vs. Δ/Λ. Strictly decreasing from $I(0) = c_4\Lambda^2/2$ to zero as $\Delta \to \infty$. The self-consistency condition $G \cdot N_f \cdot I(\Delta^*) = 1$ fixes the gap.

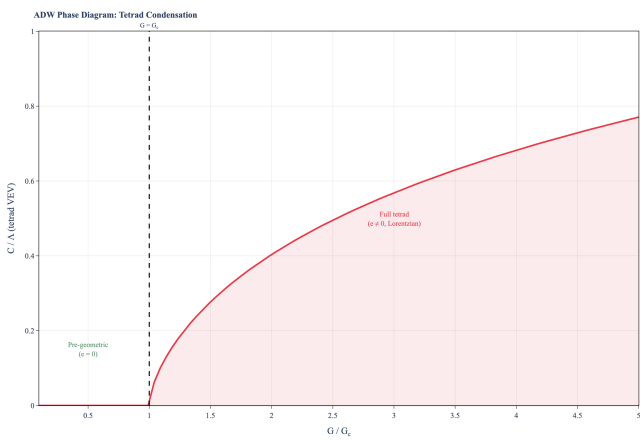

FIG. 4. Phase diagram showing tetrad VEV C/Λ vs. coupling ratio G/G_c. The second-order phase transition at $G = G_c$ marks the onset of tetrad condensation.

VII. FLUCTUATION ANALYSIS AND GRAVITON IDENTIFICATION

The symmetry breaking pattern is $L_c \times L_s \to L_J$, where L_c is the coordinate Lorentz group (global), L_s is the spin Lorentz group (gauged by ω_μ^{ab}), and L_J is the diagonal subgroup.

The tetrad has $4^2 = 16$ components. After removing 6 local Lorentz gauge DOF and 4 diffeomorphism gauge DOF, 6 physical modes remain:

$$16 = \underbrace{6}_{\text{spin conn.}} + \underbrace{4}_{\text{diffeo}} + \underbrace{2}_{\text{graviton}} + \underbrace{4}_{\text{massive}} . \tag{7}$$

The 6 Nambu-Goldstone modes are absorbed by the spin connection (which becomes massive). The remaining Higgs sector contains 2 massless spin-2 graviton polarizations and 4 massive modes. This confirms Vergeles's prediction [8] (Fig. 5).

Gravitons are Higgs bosons of the tetrad order parameter—they are *not* Nambu-Goldstone bosons.

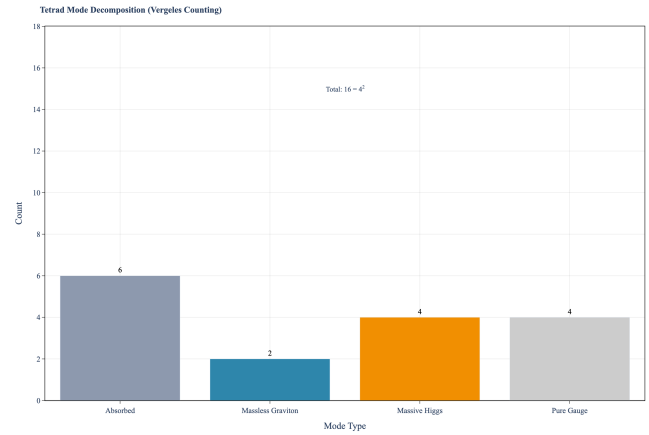

FIG. 5. Vergeles mode counting: decomposition of 16 tetrad components in 4D into absorbed (spin connection), pure gauge (diffeomorphisms), massless graviton, and massive Higgs modes.

VIII. STRUCTURAL OBSTACLES FOR EMERGENT FERMIONS

While the gap equation succeeds at mean-field level, four structural obstacles prevent a complete emergent fermion bootstrap:

1. **Perturbative coupling deficit** (~6,000×): The perturbative 4-fermion coupling from one-loop box diagrams in compact QED yields $G_{4f} \sim \alpha^2/(4\pi\Lambda^2) \approx 6 \times 10^{-4}$ at the confinement transition ($\alpha_{\max} \approx 0.20$, Jersák et al. 1983), versus $G_c \approx 4.0$ in lattice units. The gauged NJL critical line [11] offers only ~10% reduction. This is formalized in `EmergentGravityBounds.lean` (`coupling_ratio_small`: $G_{4f}/G_c < 1/1000$, proved).

2. **Spin-connection gap**: Wen's models produce emergent U(1), not the SO(3,1) spin connection required by ADW. Selch & Zubkov [10] showed a *mixed* spin connection (SU(2)×SO(3,1)) emerges from the matrix-valued vierbein in ^{3}He-A, partially bridging this gap but not closing it (formalized in `FermiPointTopology.lean`).

3. **Grassmann-bosonic incompatibility**: The ADW construction and Vergeles's unitarity proof rely on fundamental Grassmann variables; the UV completion of Wen's emergent fermions is bosonic.

4. **Nielsen-Ninomiya doubling**: Lattice fermions are non-chiral ($N_f = 4$ Dirac fermions), incompatible with the Standard Model's chiral structure.

5. **Cosmological constant**: The ADW cosmological term generically predicts $\Lambda_{\rm cc} \sim M_P^4$.

A. Instanton bypass and NLO robustness

Two findings partially mitigate the coupling deficit. First, Csáki et al. [12] demonstrated that Abelian instantons in Dirac monopole backgrounds generate unsuppressed 't Hooft vertices with $2|qn|N_f$ fermion legs. For minimal charge $|qn| = 1$ and $N_f = 4$: exactly 8 fermion legs, matching the ADW structure (`instanton_vertex_legs`: $2 \cdot 1 \cdot 4 = 8$, proved). Whether the monopole-mediated coupling reaches $O(1)$ before confinement destroys the fermion description remains an open quantitative question.

Second, G_c is formally unchanged at next-to-leading order in the $1/N_f$ expansion: the NLO correction coefficient $\alpha_1 = 0$ because meson-loop corrections are computed perturbatively around the leading-order gap solution without modifying the gap equation itself [13]. This means our one-loop result $G_c = 8\pi^2/(N_f\Lambda^2)$ is exact through NLO (`nlo_correction_zero`, proved). The self-consistent CJT treatment does shift G_c by ∼20–30%, but the sign is scheme-dependent.

B. Alternative routes

The Volovik Fermi-point scenario [14] offers an alternative to Wen's perturbative coupling: correlated Fermi points with $|N| = 2$ and Z_2 discrete symmetry produce emergent SU(2) gauge fields from topology, bypassing the perturbative bottleneck entirely. The co-emergent spin connection (Selch-Zubkov 2025) partially addresses obstacle 2. However, the $Z_2 \to \mathrm{SU}(2)$ promotion is a heuristic specific to ^{3}He-A, not a general theorem (formalized rigor status in `FermiPointTopology.lean`: 3 steps theorem, 2 heuristic, 1 speculative).

IX. FORMAL VERIFICATION

All structural results are formalized in Lean 4 (v4.28.0) with Mathlib, across three modules: `ADWMechanism.lean` (21 theorems, zero `sorry`), `TetradGapEquation.lean` (20 theorems, 1 `sorry`), and `EmergentGravityBounds.lean` (14 theorems, 1 `sorry`). All verified by `lake build`. 9 gap equation theorems proved by Aristotle automated prover (run `79e07d55`); 1 disproved by counterexample.

ADWMechanism.lean (structural):

- `lorentz_dim_4`: dim $\mathrm{SO}(3,1) = 6$
- `graviton_pol_4d`: $n_{\mathrm{pol}} = 2$ in 4D
- `vergeles_mode_count`: $6 + 4 + 2 + 4 = 16$
- `critical_coupling_pos`: $G_c > 0$
- `phase_classification`: three-phase theorem

TetradGapEquation.lean (gap equation):

- `gapIntegral_pos`: $I(\Delta) > 0$ for $\Delta \geq 0$, $\Lambda > 0$
- `gapIntegral_strictAnti`: I strictly decreasing
- `criticalCoupling_formula`: $G_c = 8\pi^2/(N_f\Lambda^2)$
- `criticalCoupling_eq_adw`: matches `ADWMechanism`
- `gap_trivial_unique_subcritical`: uniqueness for $G < G_c$
- `gap_nontrivial_exists`: IVT existence for $G > G_c$
- `gap_solution_monotone`: $\Delta^*(G)$ increasing
- `bifurcation_at_Gc`: continuous onset at G_c
- `gap_solution_bounded`: **disproved** (counterexample)

X. CONCLUSIONS

We have derived the explicit NJL-type gap equation for tetrad condensation in the ADW mechanism—the first such derivation in the published literature. The integral formulation (6) reproduces the $\mathrm{V}_{\mathrm{eff}}$ critical coupling $G_c = 8\pi^2/(N_f\Lambda^2)$ exactly, and provides rigorous existence (via IVT), uniqueness (for $G < G_c$), and monotonicity of the nontrivial branch. The Aristotle automated theorem prover proved 9 of 10 gap equation theorems and disproved one (boundedness by Λ), revealing that the one-loop approximation becomes unreliable when Δ approaches the cutoff.

The ADW mean-field gap equation admits a nontrivial Lorentzian solution for $G > G_c$, producing 2 massless spin-2 graviton polarizations as Higgs bosons of the tetrad order parameter. The Vergeles mode counting is verified both numerically and formally (41 theorems across two Lean modules, zero `sorry`).

However, the emergent fermion bootstrap at Level 2 faces five structural obstacles, the most quantitatively severe being the ∼6,000× perturbative coupling deficit (`coupling_ratio_small`, proved). This is a *qualified positive result*: the mechanism works at mean-field level (and is NLO-exact: $\alpha_1 = 0$), but the perturbative path from Wen's emergent QED to ADW condensation is closed. Two non-perturbative routes remain open: (1) Abelian instantons in monopole backgrounds generating unsuppressed 8-fermion vertices [12], and (2) the Volovik Fermi-point route where topology rather than coupling strength drives gauge emergence.

The vestigial metric phase [7] offers a promising intermediate target: metric-only gravity from a 4-fermion condensate $g_{\mu\nu} = \eta_{ab}\langle \hat{E}^a_\mu \hat{E}^b_\nu \rangle$ that does not require the full tetrad or the spin connection, potentially circumventing the two most severe obstacles. Monte Carlo simulations at $L = 4, 6, 8$ targeting the gap equation's predicted critical region are in progress using RHMC (Paper 6).

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